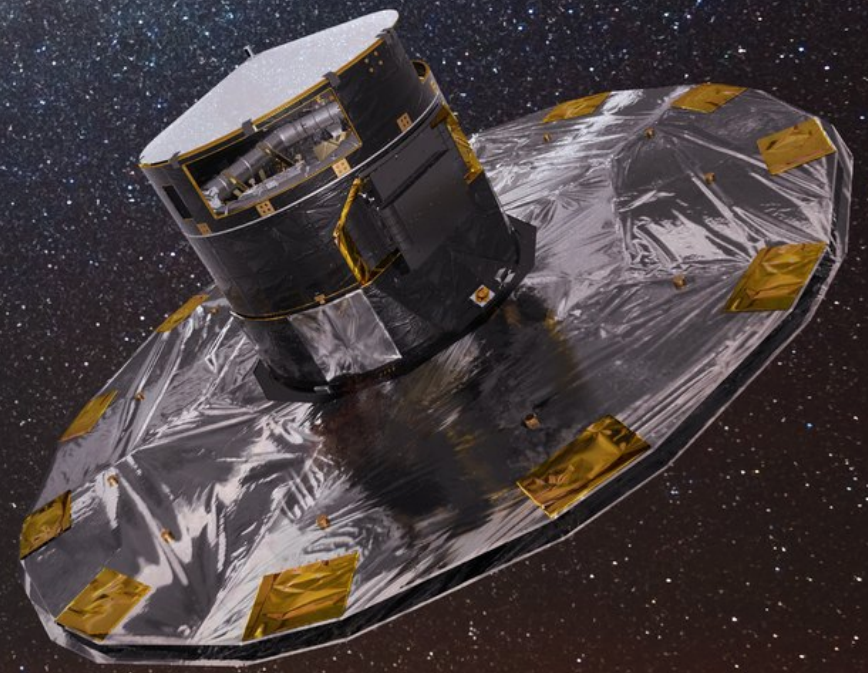
Faria · the 9 November session

Gaia

tiny angles, and a map of the Galaxy

How a European Space Agency observatory measured nearly two billion stars.



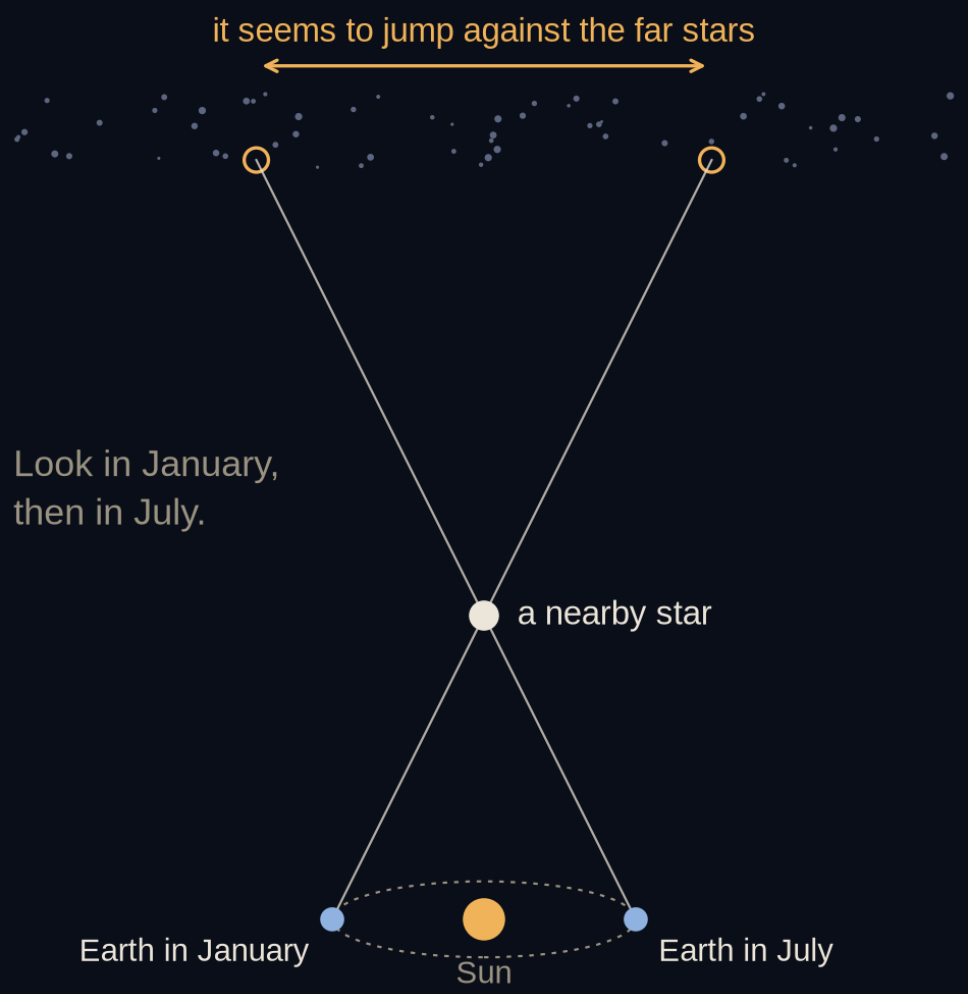
Gaia: ESA/ATG medialab; background image: ESO/S. Brunier. CC BY 4.0



Picture the Galaxy out the window

Gaia's job was to map it: where each star is, how far away, and how it moves.

A picture of the idea: looking out at the Galaxy Gaia mapped. Gaia itself carried no crew. Original teaching illustration.



What Gaia was built to measure

As Earth goes around the Sun, a nearby star seems to shift a little against the far stars.

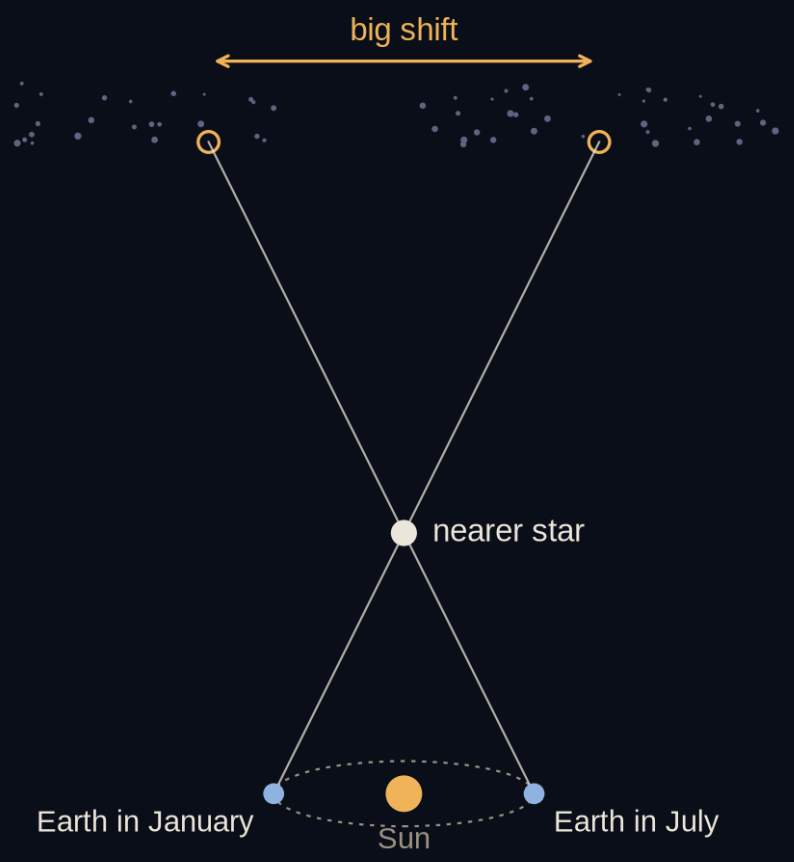
That shift is the star's parallax.

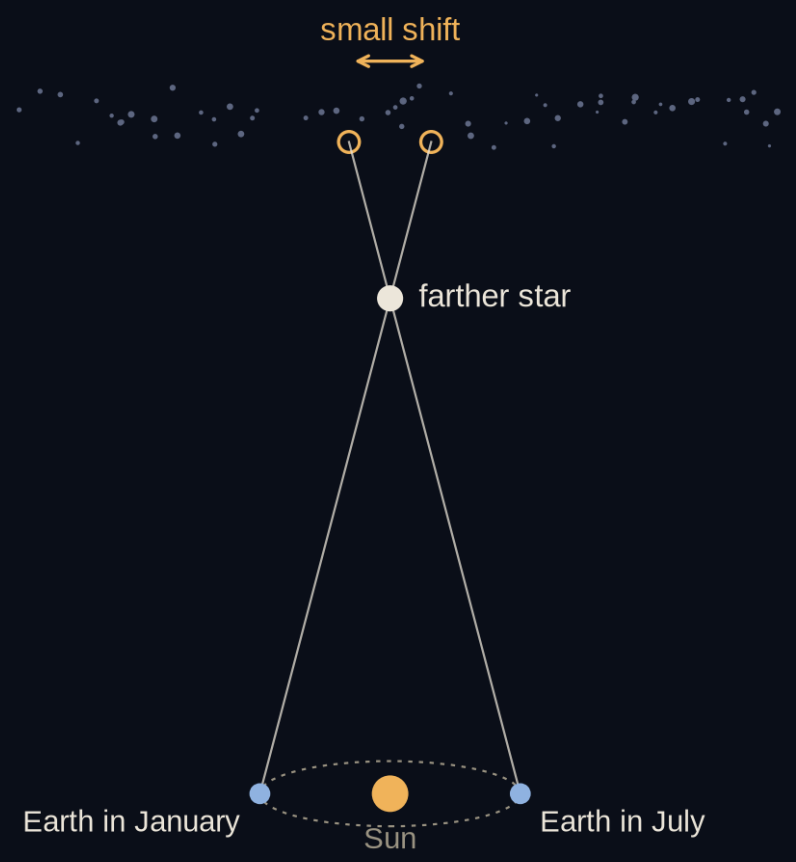
Measure the shift, and you have the distance.

Original teaching diagram. Not to scale.

The smaller the shift, the farther the star

Far stars barely move. To reach far, you must measure very small angles.





Original teaching diagram. Not to scale.

A ladder of angles

Each rung is sixty times finer: 1° = 60′, and 1′ = 60″. The sixty is Babylonian, like the sixty minutes in an hour.

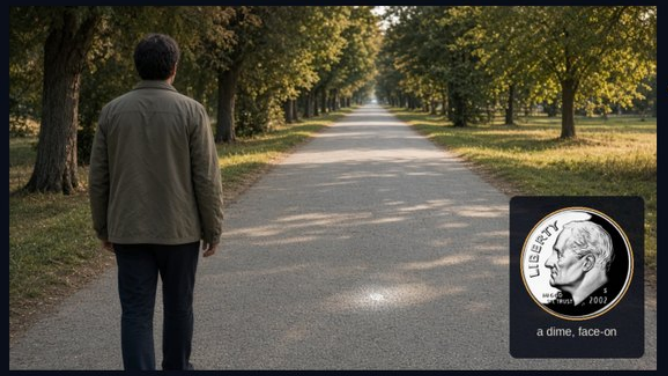
1° one degree



a pinky at arm's length

÷ 60 →

1′ one arcminute



a dime, face-on,
at about 200 feet (60 m)

÷ 60 →

1″ one arcsecond



a dime, face-on,
at about 2.3 miles (4 km)

Teaching illustrations. Dime inset: 2002 Roosevelt dime, public domain, via Wikimedia Commons.



1°

one degree

Hold out your arm. Your pinky covers about one degree of sky.

The full Moon is about half that.

Original teaching illustration.



1′

one arcminute

A dime, seen face-on, at about 200 feet (about 60 meters).

1′ = 1/60 of a degree.

Face-on means looking at the face of the dime, its full 18 mm width, never its edge. The path is a picture of the distance; at 200 feet a real dime is almost too small to see.

Teaching illustration with dime inset added. Dime inset: 2002 Roosevelt dime, public domain, via Wikimedia Commons.



1″

one arcsecond

A dime, seen face-on, at about 2.3 miles (about 4 kilometers).

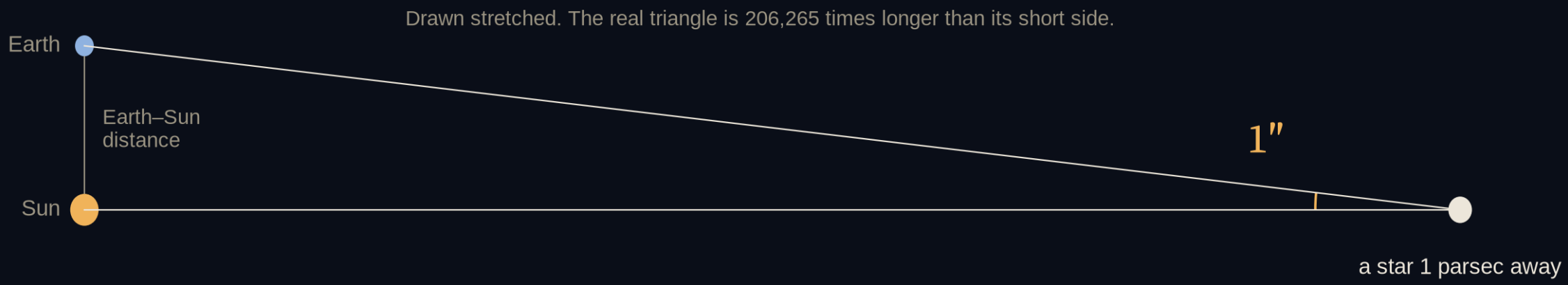
1″ = 1/60 of an arcminute, or 1/3,600 of a degree.

The same dime, face-on, sixty times farther away. The road runs about 2.3 miles to the trees on the horizon.

Teaching illustration with dime inset added. Dime inset: 2002 Roosevelt dime, public domain, via Wikimedia Commons.

One arcsecond already leaves the solar system

A star with a parallax of 1″ is one parsec away, about 3.26 light-years. Even the nearest star shifts by less than that.

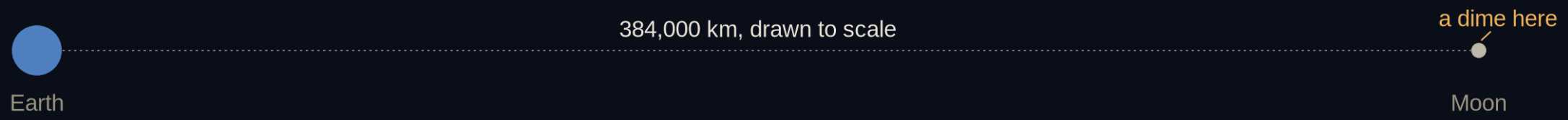


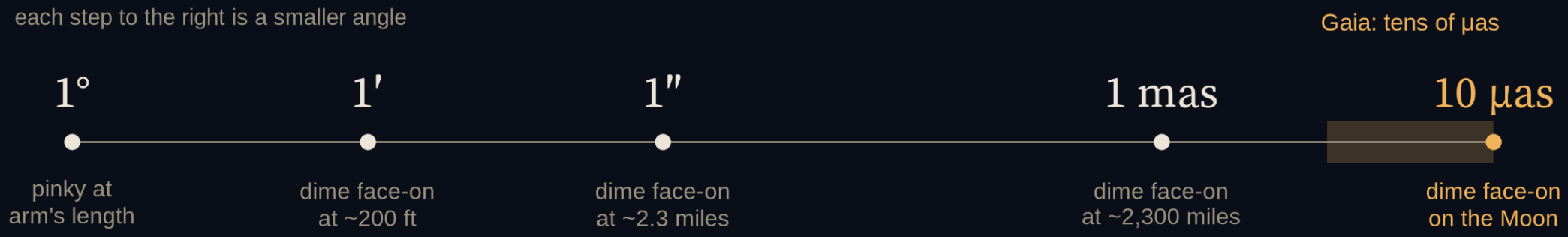


Original teaching diagram.

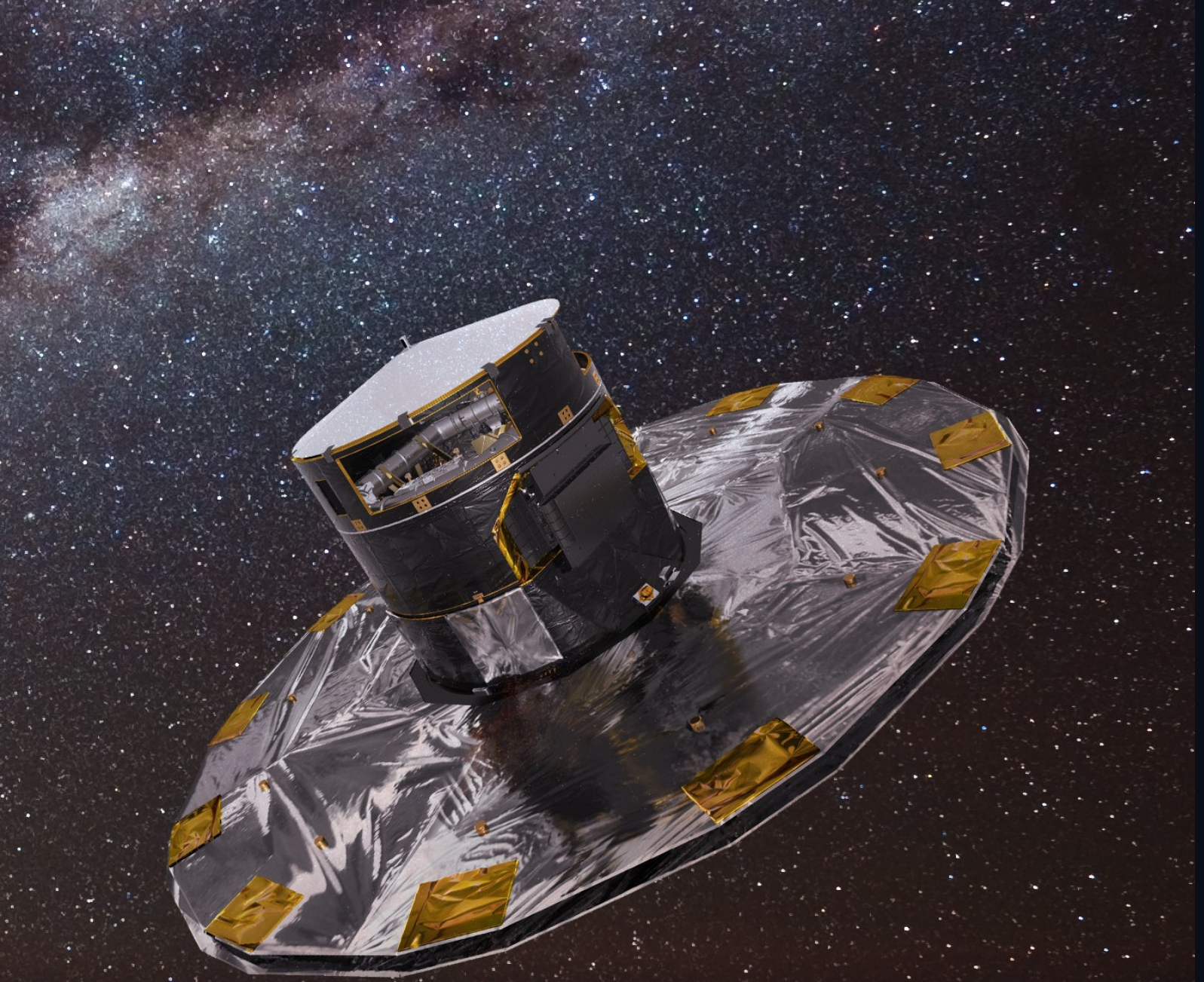
Gaia: a dime as seen from the Moon

For bright stars, Gaia measured shifts of a few tens of microarcseconds. A microarcsecond is one millionth of an arcsecond.





Original teaching diagram. A dime face-on at the Moon's distance spans about 10 μas.



Built for the European Space Agency

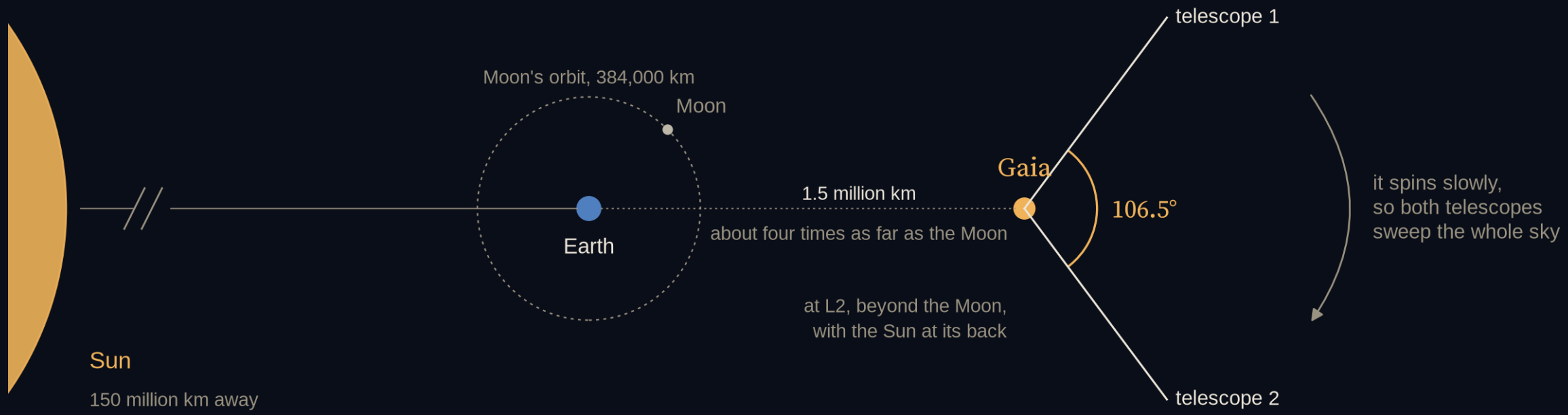
Launched on 19 December 2013. Airbus was the prime contractor.

The wide round sunshield kept the two telescopes in shade and steady.

Gaia: ESA/ATG medialab; background image: ESO/S. Brunier. CC BY 4.0

A post far beyond the Moon

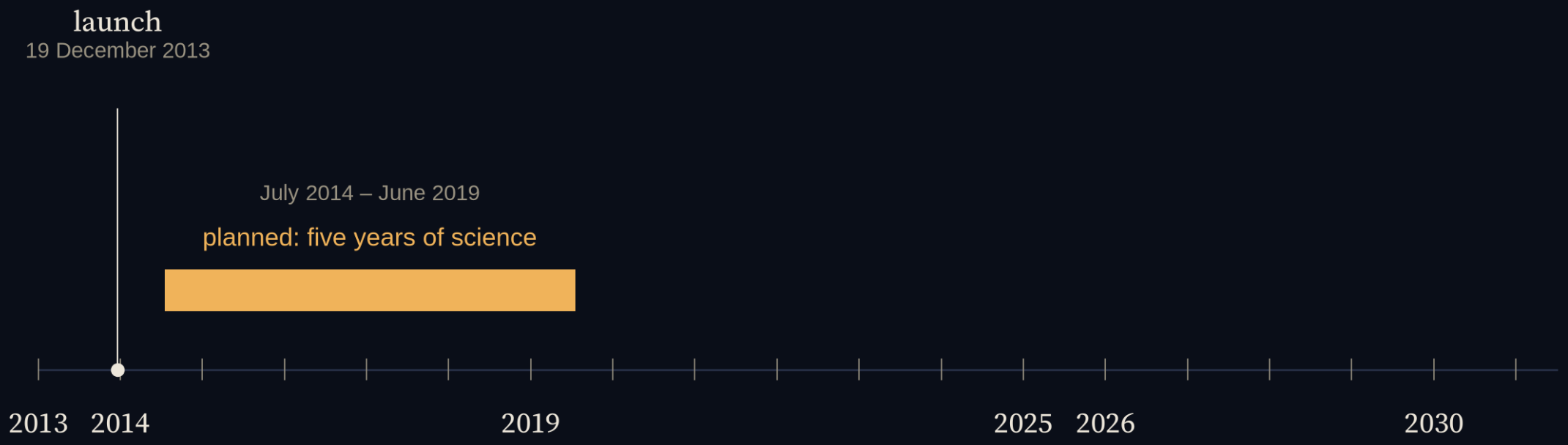
From L2, Gaia's two telescopes looked out 106.5° apart as it slowly spun, sweeping the whole sky again and again.



Original teaching diagram. Distances from Earth to scale; the Sun is not.

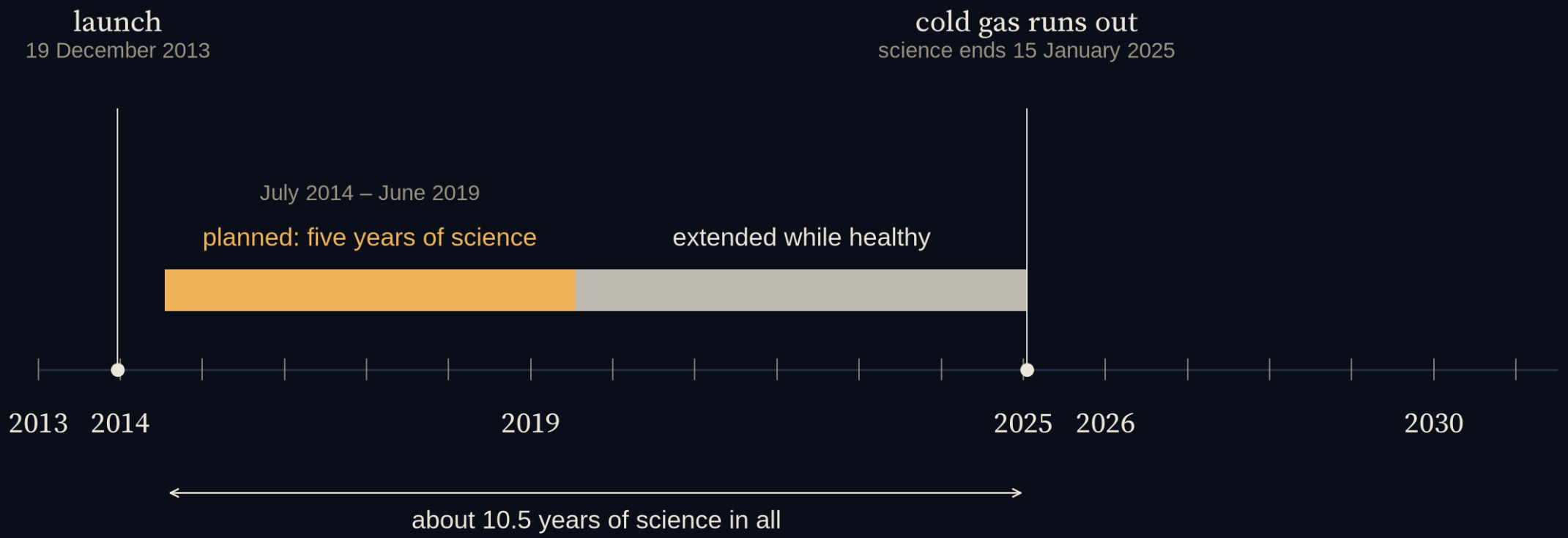
Planned for five years

Science began in July 2014 and was planned to run through the end of June 2019.



It watched for ten and a half

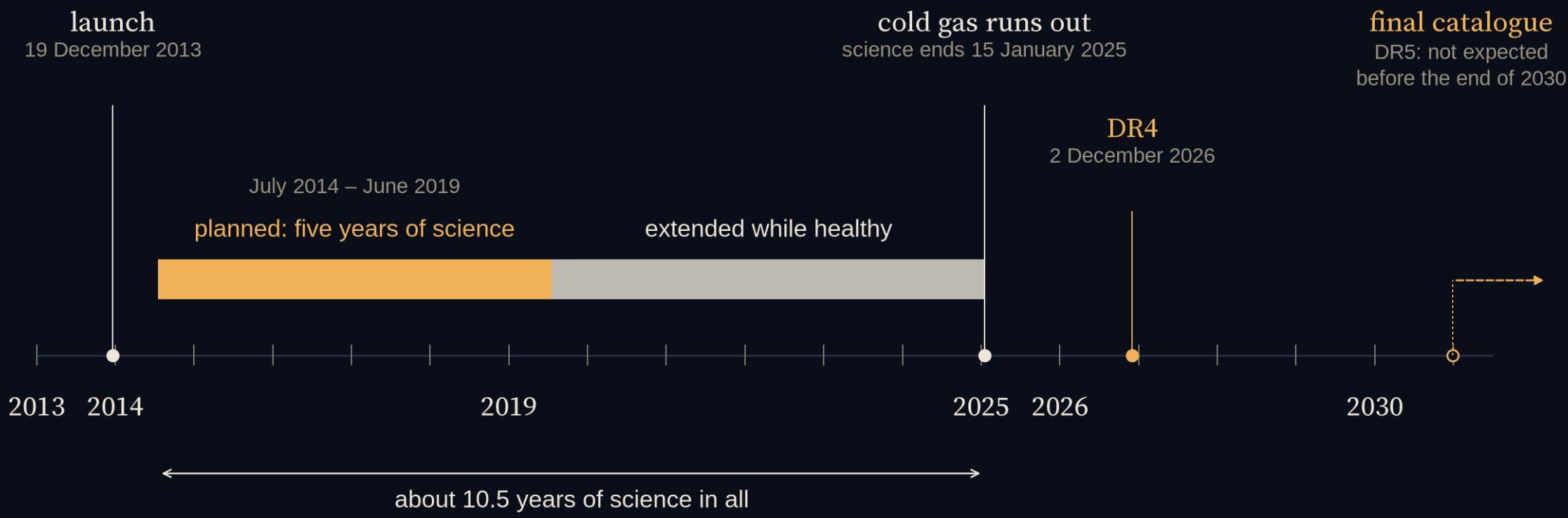
The spacecraft and the science stayed healthy, so the mission was extended, until the cold gas that kept it spinning and pointing ran out on 15 January 2025.



Switched off on 27 March 2025, and left in a retirement orbit around the Sun.

From launch to the final catalogue

Data Release 4, from the first 5.5 years, is due on 2 December 2026. The final catalogue is not expected before the end of 2030.



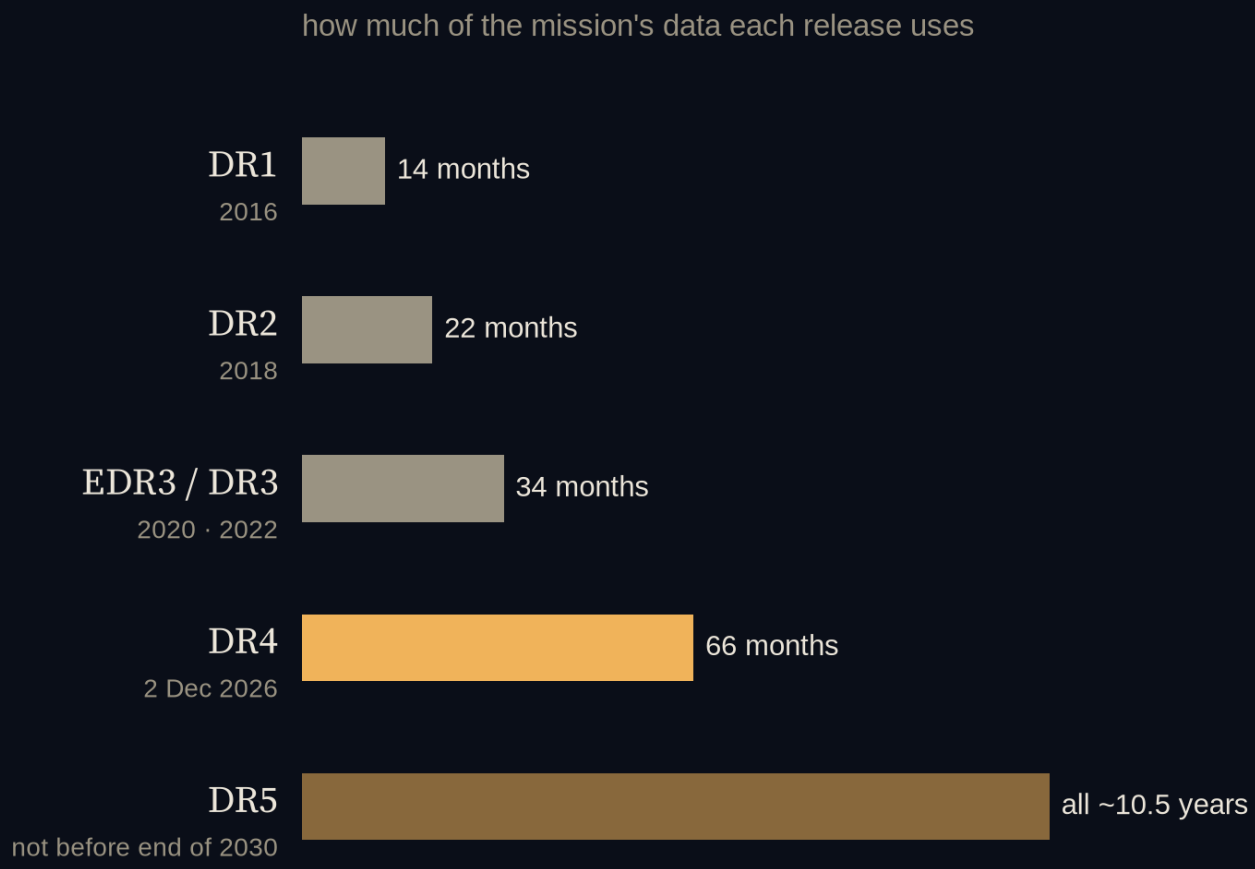
Why it takes years

3 trillion+
observations

nearly 2 billion
stars and other objects

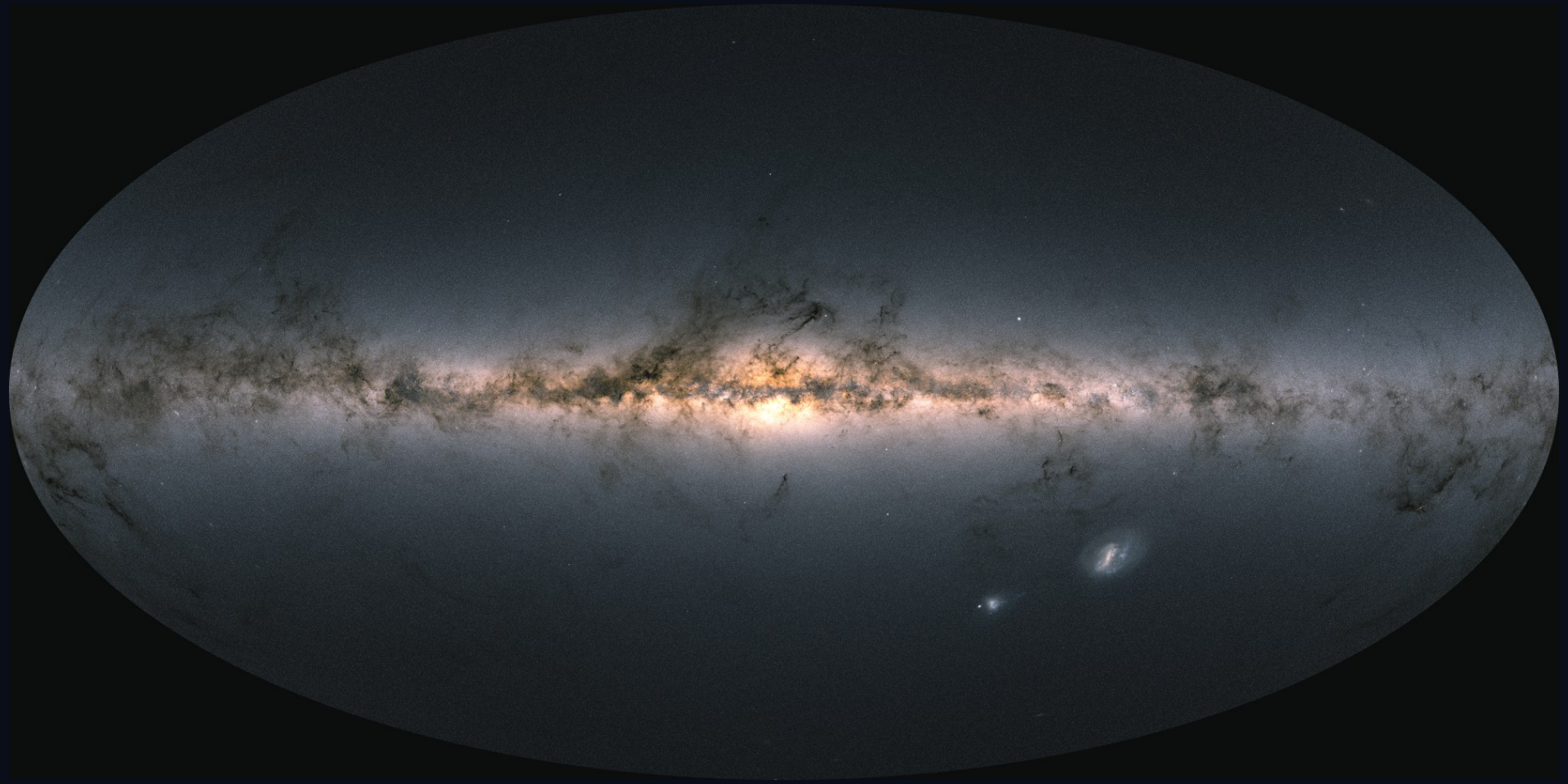
about 1,500
looks at each one, on average

Every star's position, motion and distance is solved together with the spacecraft's own pointing and calibration. Volume and complexity both take years.

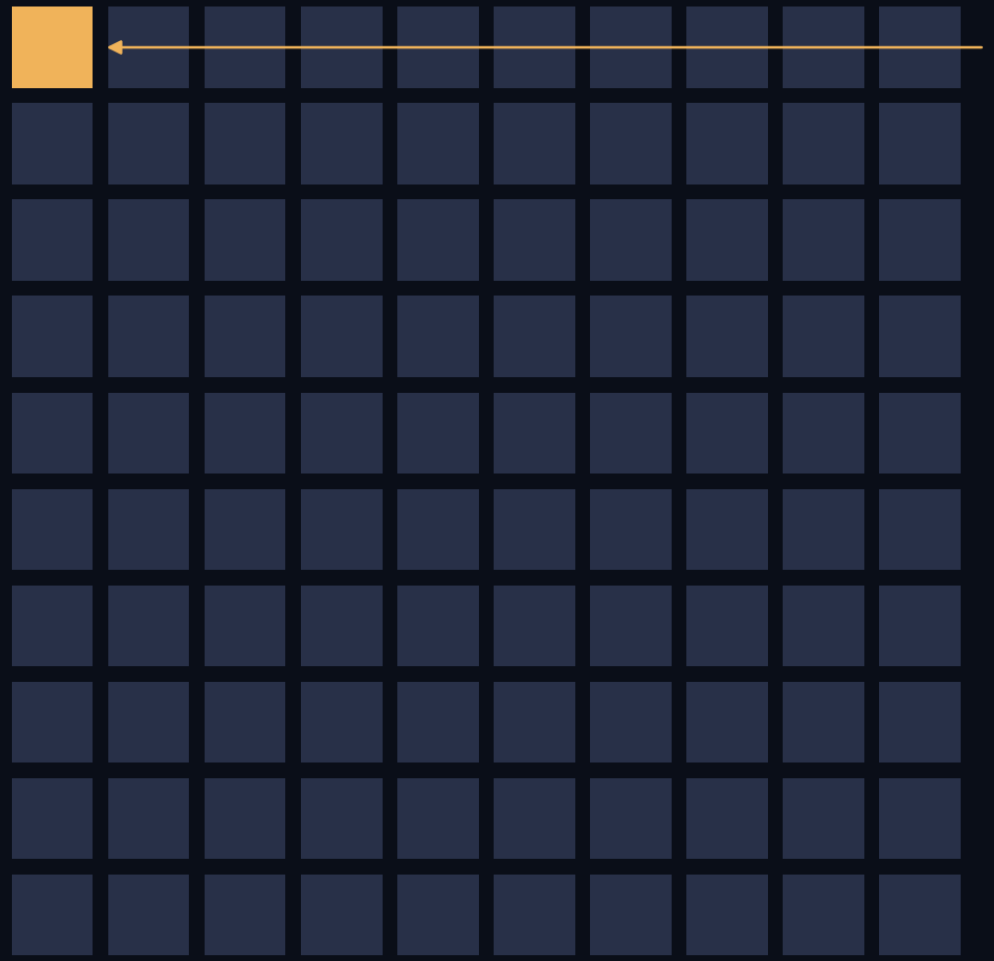


Nearly two billion stars

Gaia's own all-sky map. The bright band is the Milky Way's disk; dark lanes are dust; lower right, the two Magellanic Clouds.



ESA/Gaia/DPAC. CC BY-SA 3.0 IGO. Acknowledgement: A. Moitinho



Not every star in the Milky Way

The Galaxy holds a few hundred billion stars. Gaia measured nearly two billion, roughly one in a hundred.

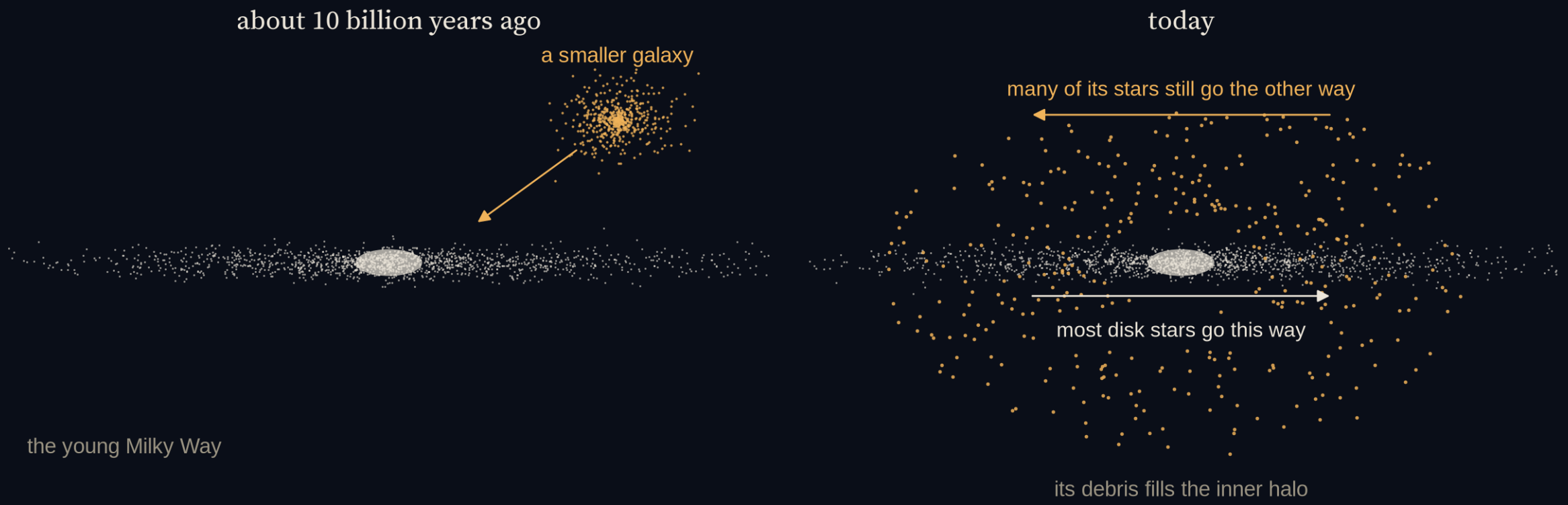
Many others are too faint, or hidden behind dust.

A huge census of the brighter stars, not a complete count.

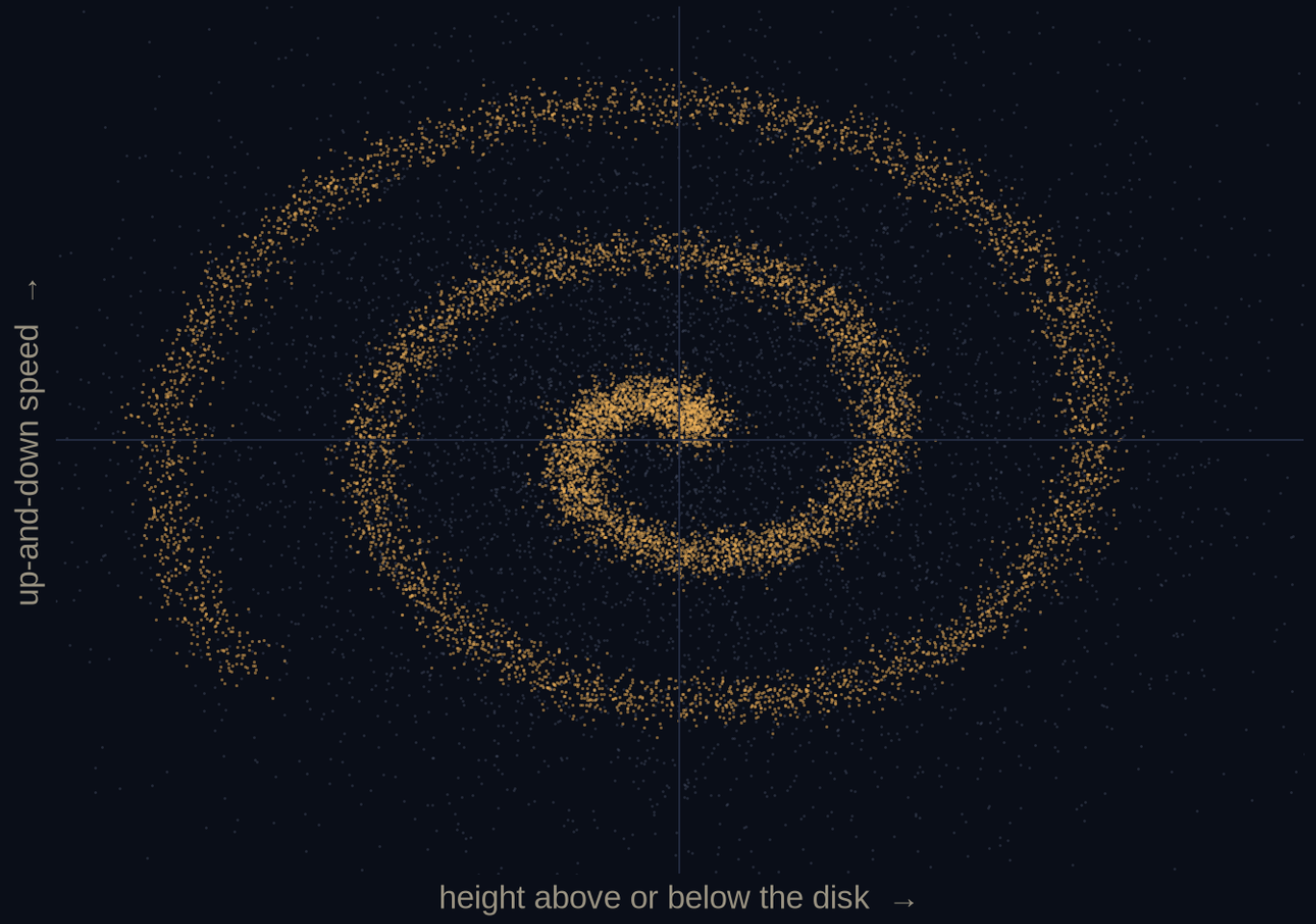
Original teaching diagram.

Gaia-Enceladus: a galaxy inside ours

In 2018 Gaia showed that much of the inner halo came from a galaxy that merged with ours about 10 billion years ago. Its stars still move like debris from the crash.



Original teaching sketch, not to scale.

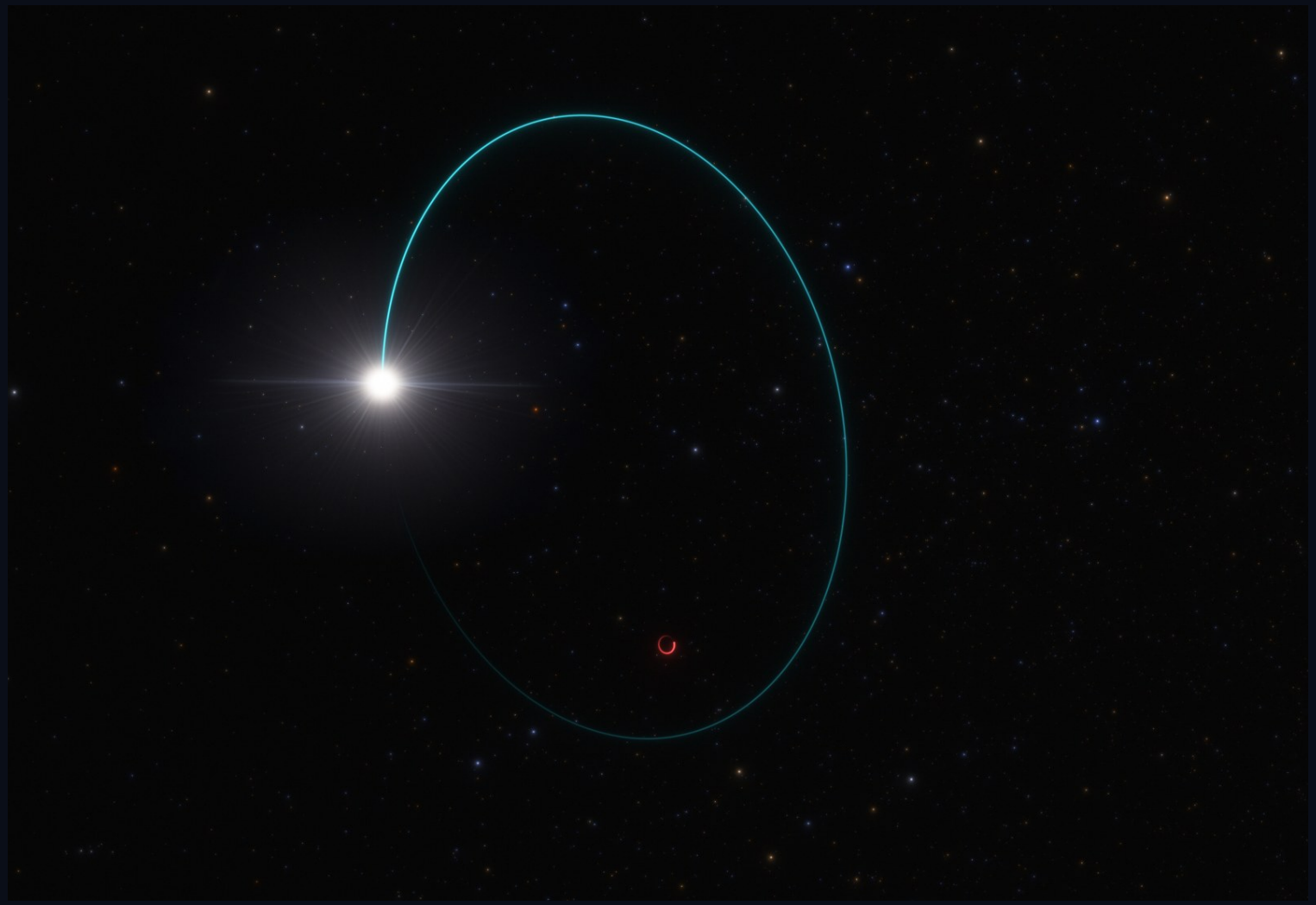


The snail-shell wobble

Also in 2018: sort nearby stars by height and up-and-down speed, and they curl into a spiral.

The disk is still ringing, like a pond after a stone, probably because the Sagittarius dwarf galaxy passed through it.

Original teaching sketch of the pattern found in Gaia DR2 (Antoja et al. 2018, Nature).



Sleeping black holes

Gaia saw stars sway around partners that give off no light.

BH1 and BH2: found in Data Release 3, confirmed in 2023.

BH3, announced 16 April 2024: about 33 times the Sun's mass, about 1,926 light-years away.

The most massive star-sized black hole then known in our galaxy.

Artist's impression of Gaia BH3: ESO/L. Calçada/Space Engine (spaceengine.org). CC BY 4.0